Zeyi Zhang

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EDUCATION

University of California, Berkeley — Berkeley, CA
Ph.D. in Chemistry (Theoretical Chemistry); candidate — *Aug 2022–2027 (expected)*
Advisors: Prof. Martin Head-Gordon and Dr. Jin Qian

Shanghai Jiao Tong University — Shanghai, China
B.S. in Chemistry, Zhiyuan Honors Program — *Sep 2018–Jun 2022*
Score: 90.64/100 GPA: 3.93/4.00 | Fan Hsu-chi Scholarship Zhiyuan Honors Scholarship

PUBLICATIONS

* Equal contribution.

[1] Xie, Y.; Shi, Z.; **Zhang, Z.**; Qi, Z.; Dong, Y.; Wang, Y.; Xiao, X. "MolP2SBench: Do LLMs Really Reason from Molecular Properties to Structure?" **Submitted to ICLR 2027** (2026). arXiv: 2610.12522 [physics.chem-ph].

[2] Li, Y.*; **Zhang, Z.***; Bai, D.; Gerstein, M. "On Policy Self Distillation for Learning Kohn-Sham Hamiltonian." **Submitted to ICLR 2027** (2026).

[3] **Zhang, Z.**; Perez, C. M.; Kwon, P.; Head-Gordon, M.; Qian, J. "PARSEC.py: A Python-Based Real-Space Kohn–Sham Density Functional Theory Code Accelerated by Machine Learned Charge Density." *J. Comput. Chem.* **47**, e70482 (2026).

[4] Dias, N.*; **Zhang, Z.***; Lemmens, A. K.; Wannenmacher, A.; Qian, J.; Head-Gordon, M.; Ahmed, M. "Ionization-Driven Oxygen–Oxygen Bond Formation in Mixed CO_2–H_2O Clusters: The Role of Solvation and Cluster Composition." *J. Am. Chem. Soc.* **148**, 35681–35693 (2026).

[5] Ling, H.; Shen, H.; **Zhang, Z.**; Head-Gordon, M. "An Energy Decomposition Method for Electronic Excitations in Two-Determinant Restricted Open-Shell Kohn–Sham Theory." *J. Chem. Theory Comput.* **22**, 385–398 (2026).

[6] Bogo, N.; **Zhang, Z.**; Head-Gordon, M.; Stein, C. J. "An Improved Guess for the Variational Calculation of Charge-transfer Excitations in Large Systems." *Phys. Chem. Chem. Phys.* **27**, 17533–17547 (2025).

[7] **Zhang, Z.**; Liu, L.; Glenna, D. M.; Jana, A.; Perez, C. M.; Qian, J. "Real-space Kohn–Sham Density Functional Theory for Complex Energy Applications." *Chem. Commun.* **61**, 10273–10286 (2025).

MANUSCRIPTS

[8] **Zhang, Z.**; Krotz, A.; Qian, J.; Head-Gordon, M. "Revisiting the Low-Lying States of the C_2 Molecule Using Hartree–Fock, Kohn–Sham Density Functional, and Non-Orthogonal Configuration Interaction Theories: Symmetry Breaking, Strong Correlation, Bonding and Energies." *Chem. Sci.* **Submitted.**

[9] **Zhang, Z.**; Qian, J.; Yao, N.; Fan, D.; Dogan, M.; Chelikowsky, J. R. "Real-Space Density Functional Theory for Materials Simulations: Algorithms, Applications, and the Next Frontier." *Phys. Rev. Mater.* **Invited contribution; in preparation.**

[10] Co-author of the next Q-Chem software overview manuscript. **Coming soon.**

EXPERIENCE

ByteDance, Seed–AI for Science · Research Scientist Intern, Quantum Chemistry — San Jose, CA
Mentors: Dr. Yixiao Chen and Dr. Xuelan Wen — *Jun 2026–Present*

- Develop **ByteQC** [Code] and **GPU4PySCF** [Code] with **tensor hypercontraction (THC)** for HF, perturbation, and CC theory, reducing average runtime and GPU memory by **38%** and **72%** on A100 GPUs.
- Develop **JaQMC** [Code] for **transition-metal** data generation via **auxiliary-field quantum Monte Carlo (AFQMC)**, yielding a Cr_2 curve closer to experiment than DMRG and **9.5%** lower BDE MAE than MRCC.
- Explore training methods for **machine-learned density functional theory (MLDFT)**.

Lawrence Berkeley National Laboratory [1, 3, 7, 9] · Graduate Student Researcher — Berkeley, CA
Advisor: Dr. Jin Qian — *Nov 2022–Present*

1. **PARSEC.py** [Code]: Develop a Python-based real-space Kohn–Sham DFT code
   - Integrate **ML density** models like ChargE3Net and SCDP, reducing SCF iterations by up to **26.7%**. [3]
   - Develop **multi-node, multi-GPU** parallelization, converging DFT for a system with **over 10,000 atoms** (∼40,000 valence electrons) in **under 5 min on 16 A100 GPUs**. [9]

- Develop GPU-accelerated geometry optimization and molecular dynamics (MD) with interfaces for **machine learning force fields (MLFF)** like UMA and MACE, reducing total relaxation time by up to **48%**. [9]
- Investigate **X-ray photoelectron spectroscopy (XPS)**, **interfacial chemistry**, and **CO_2 capture**. [7]

2. **Recursive Self-Improvement (RSI) for Science** [Demo]
   - **MolP2SBench** [1][Code]: Evaluate **7 LLMs across 11 property channels** with **shuffled and counterfactual inputs** to test whether structure predictions follow physical evidence.
   - **Paper2Bench** [Demo]: Developing benchmarks from research papers and **evaluation frameworks** to assess LLMs' ability across scientific tasks, including **reaction mechanisms, transition states, synthesis**, etc.
   - Contribute scientific tasks to **Agents' Last Exam** [Web], **FrontierPhysics** [Web], and **RSI-Exam** [Web].
3. **Digital Twin for Chemical Sciences (DTCS-v.02)** [v.01 Code] [v.01 Paper]
   - Extend DTCS, a digital twin connecting theory and experiment, with **multimodal spectroscopy** for **gas-phase reaction analysis**, targeting atmospheric and ultrafast chemistry.
   - Power DTCS with an **agentic AI workflow**, automating structure preparation, Q-Chem/VASP input generation, HPC execution, and spectroscopic analysis from natural-language instructions.
4. **Microelectronic systems**: Model semiconductor and lithography applications.

University of California, Berkeley [2, 4, 5, 6, 8] · **Graduate Student Researcher** — Berkeley, CA
Advisor: Prof. Martin Head-Gordon — *Nov 2022–Present*

1. **State-Specific Excited-State Theories, Multireference Methods, and Strong Correlation**
   - Develop **energy decomposition analysis (EDA)** for excitations in two-determinant **ROKS**. [5]
   - Develop a **frozen MO initial guess** for orbital-optimized DFT of **charge-transfer excitations**. [6]
   - Study **strongly correlated C_2** with HF, DFT, and **multireference NOCI(-RSPT)** vs. MRCI+Q/experiment, achieving **32%/24% lower bond-length/energy MAEs** than EOM-DIP-CCSD for low-lying states. [8]
2. **Computational Spectroscopy and Reaction Mechanisms**
   Study photoionization of CO_2–H_2O clusters and **O–O bond formation**, with implications for **astrochemistry**. [4]
3. **Machine-Learned Hamiltonians (MLH)**
   - Develop **label-free SCF self-distillation** for MLH, improving QH9 orbital predictions and HELM spectral stability while outperforming supervised fine-tuning across eight transfer settings. [2]
   - Develop an **electronic force field** by considering **orbital response** in non-self-consistent MLH.
   - Develop **manifold-based flow-matching models** for electronic states and molecular properties.
4. **Relativistic Quantum Mechanics and Spin–Orbit Coupling** for transition-metal complexes.

Q-Chem, Inc. [10] · **Software Engineer** — Berkeley, CA
Mentor: Dr. Xintian Feng — *Nov 2022–Present*

- Develop complex-valued orbital methods for HF, DFT, MP2, NOCI(-MP2), orbital optimization, and EDA.
- Contributed to Q-Chem 7.0, including the frozen MO initial guess for orbital-optimized DFT. [10] [Web]

NYU-ECNU Center for Computational Chemistry · **Research Assistant** — Shanghai, China
Advisors: Prof. Zenghui Zhang and Prof. Tong Zhu — *Jun 2021–May 2022*

- Built an active-learning workflow with DFT, machine learning potentials (MLP), MD, and nudged elastic band (NEB) to map pathways and transition states for 10 prebiotic glycine-formation reactions.

Shanghai Jiao Tong University · **Research Assistant** — Shanghai, China
Advisor: Prof. Huai Sun — *Feb 2021–Jun 2022*

- Investigated CO_2 solubility in ionic liquids with graph kernels and Gaussian process regression; built ILThermo data-processing pipelines for property prediction.

SKILLS

Programming & AI Agents: Python, C++, Fortran | Claude Code, Codex
Simulation Software: Q-Chem, ORCA, VASP, PySCF, Quantum Espresso, Gaussian, xTB, CREST, Molpro, ASE, CP2K, Psi4, GROMACS, LAMMPS, VMD
Scientific Computing & ML: PyTorch, JAX, CuPy, CUDA, OpenMPI, OpenMP